Module 4

Continuous-Time Dynamic Programming and Reinforcement Learning

In this document we present several MATLAB and Simulink engineering design projects relevant to dynamic programming and reinforcement learning in the continuous-time domain according to the material presented in Modules 1-4.

MATLAB Homework Problems

Homework Problem 4.1

HIV-dynamics model is defined by [1]

$$\frac{dx_1(t)}{dt} = -dx_1(t) - \beta x_1(t)x_3(t) + s$$
$$\frac{dx_2(t)}{dt} = -\mu_2 x_2(t) + \beta x_1(t)x_3(t)$$
$$\frac{dx_3(t)}{dt} = kx_2(t) - \mu_1 x_3(t)$$

where $x_1(t)$ denotes the number of healthy white blood cells (T-cells); $x_2(t)$ denotes the number of infected T-cells; $x_3(t)$ represents the number of free virus particles; $s = 10/\text{mm}^3/\text{day}$ is the source of healthy T-cells (thymus); $d = 0.02/\text{day}$ represents the death rate of healthy T-cells; $\mu_2 = 2.4/\text{day}$ represents the death rate of infected T-cells; $\mu_1 = 0.24/\text{day}$ represents the death rate of viruses; $\beta = 2.4\times10^{-5}/(\text{mm}^3\times\text{day})$ is the infectivity rate of viruses; $k = 100/\text{cell}$ is the rate of virons produced per infected T-cell. When the number of T-cells (normally $\approx 1000/\text{mm}^3$) is under $200/\text{mm}^3$, a HIV-infected patient is classified as having AIDS.

By *introducing only one virous particle*, by choosing the initial condition as $x(0) = \begin{bmatrix}1000 & 0 & 1\end{bmatrix}^T$, extremely quickly, within 20 days or so, the number of virous particles explodes to the order of 10^4. Due to the reaction of the body's immune system, these numbers will sharply decline roughly within 50 days and reach steady state values after 200 days. Similar

situation is with the number of infected T-cells. Simulate dynamics of this system. Obtain the linearized model of this system at its steady state points using the MATLAB function `linmod`.

Homework Problem 4.2

Use of mathematical modeling, optimal control systems, and reinforcement learning have attracted attention of pharmaceutical researchers and industry. Consider optimal control (drug dosing) to achieve the desired actual neutrophil counts (ANC) in chemotherapy induced myelosuppression [2]. Build the Simulink block diagram for the considered mathematical model represented by the following nonlinear fifth-order system

$$\frac{dx_1(t)}{dt} = -k_{tr}x_1(t) + k_{prol}x_1(t)\left(1 - E_{Drug}(t)\right)\left(\frac{Circ_0}{x_5(t)}\right)^{\gamma}, \quad x_1(0) = Circ_0$$

$$\frac{dx_2(t)}{dt} = k_{tr}x_1(t) - k_{tr}x_2(t), \quad x_2(0) = Circ_0$$

$$\frac{dx_3(t)}{dt} = k_{tr}x_2(t) - k_{tr}x_3(t), \quad x_3(0) = Circ_0$$

$$\frac{dx_4(t)}{dt} = k_{tr}x_3(t) - k_{tr}x_4(t), \quad x_4(0) = Circ_0$$

$$\frac{dx_5(t)}{dt} = k_{tr}x_4(t) - k_{circ}x_5(t), \quad x_5(0) = Circ_0$$

where $x_1(t)$ are proliferative cells, $x_2(t), x_3(t), x_4(t)$ are maturing cells, and $x_5(t)$ are the circulating blood cells. $E_{Drug}(t)$ stands for the chemotherapy drug effect that is a function of its concentration in the central compartment. $Circ_0$ is the base value of neutrophil count, k_{prol} is the proliferation rate constant, k_{tr} is the transit compartment rate constant, k_{circ} is the rate of neutrophil elimination, and γ is the feedback factor. The input to this nonlinear system is $E_{Drug}(t)$, and the model output denoted by $y(t)$ is the variable $x_5(t)$, that is $y(t) = x_5(t)$. The model data are taken from [2] as

$$Circ_0 = 5.04 \times 10^9 \frac{1}{\text{L}}, \quad k_{tr} = \frac{4}{120.4}\frac{1}{\text{h}} = 0.0332\frac{1}{\text{h}} = k_{prol} = k_{circ}, \quad \gamma = 0.16$$

Simulate this nonlinear system and plot the number of neutrophils during 50 days under given constant drug dosage of $250\ \text{mg}$ and the steady state drug dosage that gives a pre-specified

actual neutrophil count. Obtained the linearized model of this system at its steady state using the MATLAB function `linmod`.

References

[1] Craig, I., X. Xia, and J. Venter, "Introducing HIV/AIDS education not the electrical engineering curricula at the University of Pretoria," *IEEE Transactions on Education*, Vol. 47, 65-73, 2004.

[2] Guo, Y., N. Haddish-Berhane, H. Xie, and D. Ouellet, "Optimization of clinical dosing schedule to manage neutropenia: learning from semi-mechanistic modeling simulation approach," *Journal of Pharmacokinetics and Pharmacodynamics*, Vol. 47, 47-58, 2020.